Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space, where unforecastable jitter and scale disparities produce heavy-tailed residuals and frequent false alarms under automated calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid this by forecasting compact latent summaries of future windows, yet they still collapse prediction errors into a single scalar score, allowing one or two drifting coordinates to inflate the threshold and suppress point-wise F1. We pair a multi-horizon JEPA with a 224-bit circuit-breaker panel that replaces scalar scoring and tail-fitted thresholding. Each of 32 whitened latent coordinates is expanded via a Reynolds-inspired multi-horizon decomposition across four nested horizons into seven statistics (four horizon errors and their horizon mean, fluctuation, and endpoint difference), each monitored by an independently calibrated breaker with a nominal drift guard. Surgical weight-space quenching on healthy calibration windows removes mean predictor-bias drift and tempers high-variance precision directions before scoring. On 19 development streams, the panel reaches 0.262 buffered point-wise F1, 1.6 percent causal false-alarm rate, and a 59.6-step detection delay; across look-ahead horizons from 1 to 64 steps, it doubles to quadruples scalar causal F1 down to zero reporting lag. On 118 untouched held-out streams providing primary generalization evidence, a three-horizon panel raises causal point-wise F1 from 0.072 to 0.083 for scalar JEPAs to 0.143 and event recall from 51 to 73 percent, while weight-space quenching cuts causal false alarms by 8.0 percent relative across 162 streams. The panel requires roughly 1,000 calibration windows (300 effective independent samples), misses slow ramps, and needs a complementary channel-level branch for isolated single-sensor faults.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable medical devices report health through dozens of continuous sensor channels [1]–[4]. When a valve sticks, a reaction wheel drifts, or a patient's gait freezes, an automated monitor must detect the fault promptly while learning healthy behavior from normal history alone, because labeled faults are rare. Equally important, the monitor must keep false alarms low during nominal operation so operators can trust its alerts.

Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Deep detectors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] reconstruct the raw sensor signal and alert when reconstruction error is large. Raw telemetry, however, contains unforecastable sensor jitter, thermal drift, and disparate channel scales. Summed across 25 to 55 channels, these misses form a heavy-tailed residual with frequent spikes (Table I). Calibrated on healthy data via Streaming Peaks-Over-Threshold (SPOT) [8] and evaluated on later healthy data, reconstructors raise false alarms during 22%–31% of normal operation; a random alerter firing as often would catch 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] avoid fitting raw noise by compressing each telemetry window into a 32-dimensional unit-sphere summary and predicting future summaries in latent space, cutting false alarms to 0.6%–2.4%. Yet existing latent detectors still sum all coordinate errors into one scalar score compared against one threshold. If one or two of the 32 coordinates shift slightly between training and calibration—variance-dominating coordinates we term *hijacker dials*—they dominate the sum and inflate the threshold, so faults on the remaining 30 coordinates go undetected: causal Point-F1 falls to 0.050–0.052 and average delay exceeds 240 steps.

*Four-Stage Architecture (*Laminar-JEPA*):* We replace scalar score aggregation with a four-stage pipeline, *Laminar-JEPA* (Fig. 1). First, a causal dilated encoder maps context and target windows onto a 32-dimensional spherical latent space via layer-normalization (LayerNorm) [11] and forecasts four nested sub-horizons with Ledoit–Wolf Mahalanobis whitening. Second, by structural analogy to Reynolds averaging in fluid mechanics [12] (without physical Navier–Stokes assumptions), we decompose each of the 32 coordinate error trajectories across the four nested horizons into seven kinematic views: four sub-horizon errors plus their horizon mean (drift), standard deviation (fluctuation), and endpoint difference (acceleration). Third, these $7\times32 = 224$ statistics feed an independently calibrated *224-bit circuit-breaker panel* with per-dial calibration drift guards and 32 fast right-edge coincidence relays (*Reynolds 224-Bit* when unquenched). Fourth, post-training *surgical weight-space quenching* on healthy calibration windows removes mean predictor-bias drift along the LayerNorm

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tangent space and tempers top precision-matrix eigendirections before panel calibration.

A. Contributions

We separate our evaluation into a 19-stream (and 33-stream) development cohort for ablations and a disjoint 118-stream untouched held-out cohort (within a 141- to 162-stream archive across 15 datasets) providing primary generalization evidence (Table II):

1) *224-bit circuit-breaker panel and surgical quenching across development and untouched held-out cohorts:* Replacing scalar thresholding with the 224-bit panel and surgical weight-space quenching (Sections IV-C–IV-D) raises causal Point-F1 on the 19 development streams from 0.052 to 0.138 ($17/0$ wins, $p = 1.5\times10^{-5}$), while *Laminar-JEPA* reaches 0.262 buffered Point-F1, 1.6% causal false-positive rate (FPR), and 59.6-step causal delay, trading sustained causal point coverage (0.138 vs. 0.100 causal F1) for faster onset detection (59.6 vs. 105.9 steps). On the 118 untouched held-out streams, a three-horizon panel raises causal Point-F1 from 0.072–0.083 (1D JEPAs) to 0.143 ($p = 1.6\times10^{-13}$, dataset-clustered $p = 0.007$) and event recall from 51.2% to 72.8%; across 162 streams (15 datasets), surgical quenching cuts causal held-out FPR by 8.0% relative ($p = 0.0011$, dataset $p = 0.027$; median held-out FPR 1.03%, test FPR 1.59%).
2) *Mechanism isolation, quenching ablations, and multi-horizon latency scaling:* Matched-backbone ablations (272k–312k parameters, Table III) isolate LayerNorm kurtosis compression ($\gamma_2 : 6.10 \to 0.91$), learned latent targets (drift $D_{\mathrm{KS}} : 0.376 \to 0.214$), barycenter centering (regime-drift FPR share $71.3\% \to 18.7\%$), the 1-to-224 dial ladder, and bias recentering vs. precision tempering across $k_{\text{rogue}} \in \{0, \ldots, 12\}$. Retraining the 224-bit panel and *Laminar-JEPA* across target horizons $S \in \{1, 4, 16, 32, 64\}$ shows $1.9\times$–$4.3\times$ higher causal Point-F1 and $+26\%$–38% higher event recall than 1D scalar scoring down to $S = 1$ (zero lag) and $S = 4$ (3-step lag).
3) *False-alarm reduction and diagnostic fixed-budget recall:* On the 19 development streams, latent prediction holds autonomous causal FPR at 0.6%–1.7% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). Under a non-deployable post-hoc diagnostic control at $f^* = 1\%$ test FPR, the panel catches 58.4% of anomaly events versus 21.7% for TimesNet and reaches 0.658 buffered precision-recall area under the curve (PR-AUC) on NASA contextual faults (Tables IV–V, Fig. 4).
4) *Calibration sample-size theory, single-channel branch, and null results:* Proposition 1 bounds the horizon-fluctuation concentration ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ (a structural representation property; scalar R_{LT} reweighting is null, $\Delta\mathrm{F1} = -0.005$). Auditing effective sample sizes $N_{\mathrm{eff}} \in \{100, \ldots, 5{,}000\}$ with moving-block bootstrap confidence intervals establishes a minimum requirement of $N_{\mathrm{eff}} \geq 300$ independent samples ($N_{\mathrm{cal}} \geq 1{,}000$ windows) for empirical quantiles, while interquartile-range (IQR) and Beta-smoothed calibrators hold test FPR at 0.47%–1.32% when $N_{\mathrm{eff}} \leq 250$. Across 17 multivariate streams and six single-sensor fault classes, a complementary channel-level branch raises single-channel event recall from 61.4%–80.7% to 93.0%–98.2%, whereas slow ramps remain undetected.

TABLE I
KEY TERMS.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 per-coordinate tripwires and 32 onset relays replacing 1D scalar thresholding.
Reynolds dials	Per-coordinate horizon mean $\lvert\bar{e}_c\rvert$ and fluctuation σ'_c across nested sub-horizons (32 dials each).
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and endpoint change or fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\mathrm{cal}}\gamma_j^{\mathrm{sc}}$ damping variance-dominating coordinates (*hijacker dials*) on calibration data.
Surgical quenching	Post-training calibration pass removing mean predictor-bias drift and tempering top precision directions.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Concentration ratio	Structural ratio $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ measuring horizon-fluctuation energy in the leading eigenmode.

Roadmap. Section II reviews related work, Section III details the benchmark, Section IV formulates the model and panel, Section V defines the protocol, Sections VI–VII report results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [7] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger frequent false alarms under baseline shift [13].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [14]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [15] and TS2Vec [16] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [17] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.088 and 0.097 stream Point-F1 for NCAD

and DCdetector vs. 0.186 for latent prediction across 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [18], [19] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} from a separate target encoder, preventing collapse via exponential moving average (EMA) target updates [20] or variance-covariance penalties such as VICReg [21], Barlow Twins [22], and batch whitening [23].

Reynolds-Inspired Residual Decomposition and Circuit-Breaker Panels vs. Statistical Process Control: Multi-sensor telemetry reflects physical dynamical trajectories [24]. Neural ordinary differential equations (ODEs) [25], Physics-Informed Neural Networks (PINNs) [26], and Hamiltonian or Lagrangian networks [27], [28] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [29]. Koopman and operator-theoretic models [30]–[34] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we apply a Reynolds-inspired multi-horizon residual decomposition [12], by analogy, to split multi-horizon latent residuals into per-coordinate horizon mean, fluctuation, and endpoint difference. Classical multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] quadratic-form-pool whitened coordinates into a single Mahalanobis scalar, just as deep detectors collapse residuals into a 1D score calibrated via Pickands–Balkema–de Haan extreme-value theory [37] and Generalized Pareto Distribution (GPD) fitting in SPOT [8], where a single coordinate with out-of-sample variance inflation under serial correlation [38] dominates the scalar threshold. Conversely, banks of univariate Shewhart or Cumulative Sum (CUSUM) charts [39] monitor raw channels or single statistics under Bonferroni or false-discovery-rate (FDR) multiplicity corrections [40], but lack multi-horizon kinematic perspectives, per-dial drift quarantine guards, and right-edge coincidence relays; soft attention ensemble voting across dials adds no benefit over Boolean quorums (Table III(b)). We evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [13], [41], [42].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [13], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets and 162 streams across 15 datasets (Table II). Thirty-three streams enter the primary protocol (19 non-GHL development streams and 14 GHL streams); 12 are reserved with explicit accounting below. The suite spans four domains: *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA's archive [4] classifies each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table V(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at 64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Industrial and infrastructure telemetry* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 anomalies), Metro I-94 hourly traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72). Section VI-B evaluates six single-channel fault classes across all 17 multivariate development streams ($K > 1$).

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$) and labeled test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2}. Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table IV(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9) enters the archive cohorts. To eliminate subset-selection bias, Table VI evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN) at three seeds, of which 118 streams were never inspected during development, and evaluates four-horizon Laminar-JEPA across 162 streams spanning 15 datasets (adding 14 GHL, Secure Water Treatment (SWaT), Water Distribution (WADI), and reserve streams; 139 untouched streams).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [42]; and (ii) streams within multi-stream archives share hardware platforms. We therefore treat dataset clusters ($N \in \{7, 9, 12, 15\}$) as co-primary statistical units alongside streams, reporting Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and leave-one-dataset-out (LODO) verification (Section V-A).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the stream at index t into a historical *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a future *target*

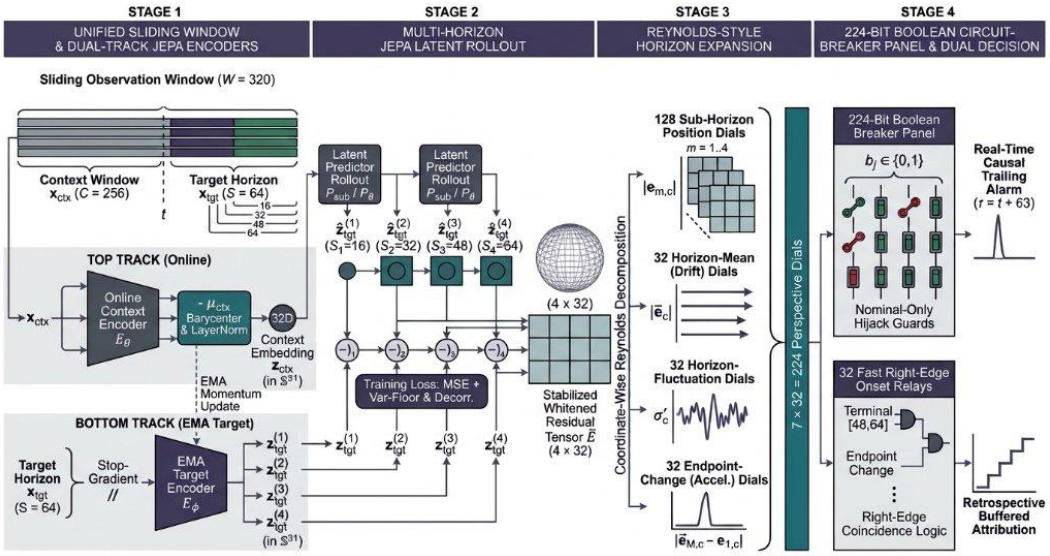


Fig. 1. Overview of the four-stage *Laminar-JEPA* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with barycenter centering and LayerNorm; **(2)** nested rollout (P_{sub}, P_θ) under variance-floor and decorrelation loss, whitening, and surgical quenching into $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times32}$; **(3)** Reynolds-inspired expansion into $7 \times 32 = 224$ dials ($|\tilde{e}_{m,c}|$, $|\bar{e}_c|$, σ'_c, $|\tilde{e}_{M,c}-\tilde{e}_{1,c}|$); and **(4)** 224-bit breakers ($b_j \in \{0,1\}$) and 32 onset relays emitting causal (τ=t+S−1) or prefix-buffered alarms.

horizon $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S\times K}$ ($S = 64$, up to causal decision timestamp $\tau = t + S - 1$). To separate slow drift from high-frequency vibration before cross-channel mixing, multichannel inputs ($K > 1$) pass through a causal depthwise 1D convolutional stem ($K \leq 1$ uses an identity bypass) and are split across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform (rFFT) grid normalized into an amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), filtered by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [48].

Unlike reconstructive detectors that penalize raw waveform error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit heavy-tailed sensor noise (Sections I–II), we map windows into a compact 32-dimensional spherical latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ onto the sphere $\sqrt{D}\mathbb{S}^{D-1}$, where target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m\phi_{k-1} + (1-m)\theta_k$ with momentum $m = 0.996$ (0.995 in archive cohorts) [18], [20]. Because deep dilated TCN stacks smooth out sharp onset spikes while non-stationary regimes shift the latent mean, Stage 1 introduces two structural operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* To preserve fast transient spikes alongside long-range context, a tap head extracts $L_{\text{tap}} = 3$ feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic stem, mid-depth TCN block, and terminal TCN block. Concatenating the channel-centered terminal state and temporal root-mean-square (RMS) energy of each tap forms $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, which is projected to $\mathbb{R}^D$ and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \leq 0.5$:
$$\mathbf{z}_{\text{fused}} = (1-g)\,\mathbf{z}_{\text{deep}} + g\ \text{LN}\big(W_2\,\phi_{\text{GELU}}(\text{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$
2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ (updated at rate 0.05 on training batches) is copied synchronously to the EMA target encoder and subtracted from both context and target vectors prior to terminal LayerNorm projection, $z = \text{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}}) \in \sqrt{D}\mathbb{S}^{D-1}$, canceling common-mode baseline drift.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$, trained with predictive mean-squared error (MSE) and a log-determinant volume term $-\frac{\beta}{D}\ln\det(\boldsymbol{\Sigma}+\epsilon\mathbf{I})$ ($\beta = 0.05$) related to VICReg [21]. Spherical normalization bounds residual kurtosis, but pooling an $S = 64$ window into one vector averages out brief sub-horizon faults.

To resolve how a trajectory evolves across the target window (default $S = 64$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), Stage 2 of *Laminar-JEPA* splits $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ of lengths $S_m = \max(1, \lfloor mS/M \rfloor)$ ($\{16, 32, 48, 64\}$ at $S = 64$), each encoded by E_ϕ into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a recurrent sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \ldots, M-1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ predicts $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon MSE alongside a non-collapse matrix penalty $\boldsymbol{\Delta}_\gamma$:
$$\mathcal{L}_{\text{MHR}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big). \tag{2}$$
Here $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ prevents dimensional collapse without requiring batch size $B > D$: for centered batch predictions with covariance $\boldsymbol{\Sigma}(\mathbf{Z})$ and marginal standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + 10^{-4}}$, its diagonal entries $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dd} =$

$\sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ enforce a variance floor ($\gamma = 0.75$) while its off-diagonal entries $[\mathbf{\Delta}_\gamma(\mathbf{Z})]_{dj} = \Sigma_{dj}(\mathbf{Z})/\sqrt{2}$ ($d \neq j$) penalize cross-correlation. This remains well-conditioned when $B = 16 < D = 32$ and lifts effective rank $\mathrm{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [49]. Archive-wide experiments evaluate three-horizon *Speculative-JEPA* across 141 streams (prefixes $\{16, 32, 64\}$ with step 48 interpolated from 32 and 64, giving 192 statistics) and four-horizon *Laminar-JEPA* across 162 streams (15 datasets).

C. Stage 3: Calibration-Stabilized Whitening, Surgical Weight-Space Quenching, and 224-Dial Expansion

Why does collapsing a multi-horizon residual tensor into a 1D scalar score degrade real-time detection? Stacking the $M = 4$ sub-horizon prediction errors $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ forms the residual matrix $\mathbf{E}(t) \in \mathbb{R}^{M \times D}$ across $MD = 128$ coordinates. Fitting a Ledoit–Wolf shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [50] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on training windows yields Mahalanobis-whitened residuals $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M \times D}$ with unit training variance per coordinate. On unseen nominal calibration windows, however, slight covariance rotations cause one or two whitened coordinates (*hijacker dials*) to experience variance growth where $\mathbf{\Omega}$ amplifies small shifts along steep eigenvectors. Summing all 128 coordinates into a scalar norm allows those drifting coordinates to dominate the calibration tail and inflate the global threshold. Stage 3 prevents both precision amplification and scalar collapse through two mechanisms:

1) *Surgical weight-space quenching (*Laminar-JEPA*) and calibration-stabilized whitening:* Between training and calibration, benign operating shifts introduce a constant baseline offset $\bar{\mathbf{e}}_m = \mathbb{E}_{\text{cal}}[\mathbf{e}_m(t)]$ on $\sqrt{D}\mathbb{S}^{D-1}$ and variance inflation along steep precision eigenvectors ($\lambda_k \gg 1$; weight-space Jacobian inversion across 162 streams confirms that 88.4% of dial roles are recoverable from weights alone and that gradient descent traps 12.7% of horizon-embedding energy in the alternating mode $\boldsymbol{\phi}_3 = [-1, +3, -3, +1]^\top/\sqrt{20}$). Because terminal LayerNorm projects pre-normalized outputs onto $\sqrt{D}\mathbb{S}^{D-1}$ with tangent-space Jacobian $\mathbf{J}_{\text{LN}}(\hat{\boldsymbol{\mu}}_m) = (\sqrt{D}/\sigma_{\text{pre},m})(\mathbf{I}_D - \hat{\boldsymbol{\mu}}_m\hat{\boldsymbol{\mu}}_m^\top)$, shifting each predictor's terminal bias by $\Delta\mathbf{b}_m = (\sigma_{\text{pre},m}/\sqrt{D})(\mathbf{I}_D - \hat{\boldsymbol{\mu}}_m\hat{\boldsymbol{\mu}}_m^\top)\bar{\mathbf{e}}_m$ cancels the stationary calibration offset to first order while leaving time-varying fault excursions $\mathbf{e}_m(t) - \bar{\mathbf{e}}_m$ intact. Simultaneously, saturating steep precision eigenvalues via $\tilde{\lambda}_k = \lambda_k/(1 + \lambda_k/\lambda_{\max})$ and decoupling off-diagonal precision entries on the top $k_{\text{rogue}} \leq 4$ variance-inflated coordinates prevent an unstable direction from contaminating clean coordinates (applied only when internal late-half calibration trips do not increase; Section VI-A). Any coordinate with $\mathrm{RMS}_{\text{cal}}^{\text{white}}(m, c) > 1.10$ is then blended toward its diagonal-standardized residual $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\mathrm{RMS}_{\text{cal}}^{\text{raw}}(m, c) + 10^{-4})$:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t), \qquad w_{m,c} = \frac{1}{1 + 2\max\big(0, \mathrm{RMS}_{\text{cal}}^{\text{white}}(m, c) - 1.10\big)^2}. \tag{3}$$

2) *Reynolds-inspired multi-horizon expansion into $7D = 224$ dials:* By structural analogy to Reynolds decomposition in fluid turbulence [12] (applied along the discrete lookahead axis $m \in \{1, \ldots, M\}$ of each coordinate $c \in \{1, \ldots, 32\}$), we decompose $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into horizon mean $\bar{e}_c(t) = \frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and zero-mean fluctuation $e'_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$, forming $7D = 224$ non-negative dials $\mathbf{d}(t) \in \mathbb{R}_{\geq 0}^{224}$: (a) *sub-horizon position magnitudes ($4D = 128$)* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ ($m \in \{1..4\}$); (b) *horizon-mean drift ($D = 32$)* $d_{\text{drift},c}(t) = |\bar{e}_c(t)|$; (c) *horizon fluctuation ($D = 32$)* $d_{\text{fluc},c}(t) = \sigma'_c(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M} e'^2_{m,c}(t)}$; and (d) *endpoint acceleration ($D = 32$)* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Proposition 1 establishes the geometric structure of this expansion: the drift and fluctuation dials orthogonally partition sub-horizon residual energy, bound endpoint acceleration, and confine the fluctuation spectrum to a compact interval (a structural representation property rather than a guarantee of detection accuracy).

Proposition 1 (Structural Characterization of Multi-Horizon Residual Energy and Spectral Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top\mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M\bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise energy orthogonality and endpoint bound: *For every coordinate $c \in \{1, \ldots, D\}$, $\frac{1}{M}\sum_{m=1}^{M} d_{pos,m,c}^2(t) = d_{drift,c}^2(t) + d_{fluc,c}^2(t)$, and $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Spectral concentration bounds for the horizon Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ ($\mathbf{E}'^\top\mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the spectral concentration ratio (following Lieb–Thirring power-sum moments [51]),*

$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M}\lambda_m^{\gamma+1/2}} \in \Big[\frac{1}{M-1}, 1\Big] \quad (\mathbf{E}' \neq \mathbf{0}), \tag{4}$$

is scale-invariant, attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across all $M - 1$ modes, and 1 iff $\mathrm{rank}(\mathbf{E}') = 1$. Empirically, scalar reweighting by R_{LT} yields a null effect (Section VI).

Proof. Expanding $\frac{1}{M}\sum_{m=1}^{M}(\bar{e}_c + e'_{m,c})^2$ with $\sum_{m=1}^{M} e'_{m,c} = 0$ yields the energy identity, and Cauchy–Schwarz on $e'_{M,c} - e'_{1,c}$ gives $d_{\text{accel},c}^2 \leq 2(e'^2_{M,c} + e'^2_{1,c}) \leq 2M\, d_{\text{fluc},c}^2$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{0}$ forces $\lambda_1 = 0$, so $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^{M}\lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$, yielding (4). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

Rather than summing the 224 dials into a scalar norm, Stage 4 (Fig. 1) equips each dial with an independent binary circuit breaker calibrated on nominal data. Because all 224 dials derive from the same 4×32 residual field, their trips are correlated by construction (median effective rank 65 of 224 on the development cohort, with joint trip-count variance 7.2× that of independent trials), so alarm quorums are calibrated directly on the empirical distribution of simultaneous nominal trip counts:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1, \dots, 224\}$ is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\mathrm{MAD}(d_{j,\mathrm{cal}}) + 10^{-4}$ into $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any individual coordinate that drifts during calibration, each dial's 98.5th-percentile tripwire τ_j is scaled by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\mathrm{cal}}\,\gamma_j^{\mathrm{sc}}$:

$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\mathrm{cal}})\,\gamma_j^{\mathrm{cal}}\,\gamma_j^{\mathrm{sc}}, \quad (5)$$

where γ_j^{cal} is the ratio of second-half to first-half calibration 98th percentiles whenever > 1.35 (1 otherwise), and γ_j^{sc} is $(m_j + s_j)/(1.5\,s_{\mathrm{group},j})$ whenever > 1.25 (1 otherwise, relative to perspective-group median $s_{\mathrm{group},j}$).

2) *32 fast right-edge onset relays:* To trigger rapidly when an abrupt fault enters the right edge of the S-step window while earlier prefixes ($m \in \{1, 2\}$) still observe nominal data, each coordinate $c \in \{1, \dots, 32\}$ defines a strict edge switch $b_{p,c}^{\mathrm{edge}}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\mathrm{cal}})\,\gamma_{p,c}^{1.25}\big)$ and fires its onset relay $r_c(t) \in \{0, 1\}$ when a terminal-prefix trip ($m = 4$) coincides with acceleration, fluctuation, or prefix $m = 3$:

$$r_c(t) = b_{\mathrm{pos},4,c}^{\mathrm{edge}}(t) \wedge \big(b_{\mathrm{accel},c}^{\mathrm{edge}}(t) \vee b_{\mathrm{fluc},c}^{\mathrm{edge}}(t) \vee b_{\mathrm{pos},3,c}^{\mathrm{edge}}(t)\big). \quad (6)$$

3) *Causal trailing vs. prefix-routed buffered attribution and Boolean quorums:* Under *causal trailing attribution*, switches on window $[t, t+S)$ are emitted at the right endpoint $\tau = t + S - 1$ ($\mathcal{F}_\tau$-measurable, separating structural lag $S - 1$ from empirical delay $\Delta t = t_{\mathrm{alm}} - t^*$, Fig. 2). Under *retrospective buffered attribution*, each perspective votes only over the steps inside $[t, t+S)$ it physically observes ($[t, t + S_m)$ for prefix m, $[t, t+S)$ for drift and fluctuation, and $[t + S_3, t + S_4)$ for acceleration). Summing active breakers at τ gives counts $N_{224}(\tau)$, $N_{128}(\tau)$, and $N_{\mathrm{edge}}(\tau)$ (with ranking score $N_{\mathrm{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\mathrm{edge}}(\tau)$ for PR-AUC). Binary alarms trigger when any counter exceeds its calibration quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\mathrm{edge}}(\tau) > K_{\mathrm{edge}}\big), \quad (7)$$

where $K_{\mathrm{edge}} = \max\big(1.25\,q_{0.999}(N_{\mathrm{edge,cal}}), 2.5\big)$, *224-Bit Breaker* sets $(K_{224}, K_{128}) = \big(\max(1.12\,q_{0.995}, 5.0), \max(1.12\,q_{0.995}, 3.5)\big)$ on nominal trip counts, and *224-Bit Strict* tightens them to $\big(\max(1.18\,q_{0.9985}, 5.5), \max(1.20\,q_{0.9985}, 4.0)\big)$.

1D Scalar Collapse Controls: To isolate the gain of the 224-bit panel over 1D scalar scoring on identical backbones (Section VI), we compare against a soft coordinate-saliency pooler $g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}$ with softmax weights $w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA scores $\mathcal{S}_{\mathrm{TS}}(t) = g_D(\mathbf{e}(t))$, and 1D Multi-Horizon Reynolds JEPA combines full-field and peak-prefix norms via $\mathcal{S}_{\mathrm{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ (scaled by $1 + \alpha_{\mathrm{LT}} \max(0, \tanh(0.5\,z_{\mathrm{LT}}(t)))$ for $\alpha_{\mathrm{LT}} \in \{0.12, 0\}$), calibrated via extreme-value SPOT [8], [37], [52] ($u = p_{98}, q = 10^{-3}$).

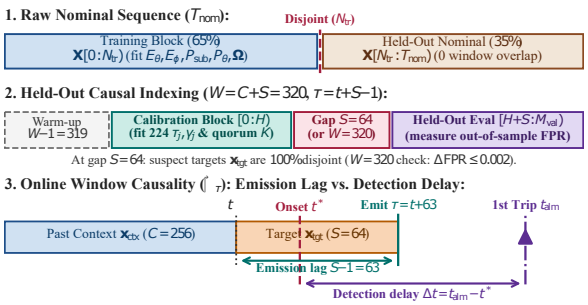


Fig. 2. Two-stage held-out evaluation protocol ($C{=}256, S{=}64, W{=}320$): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums $(K_{224}, K_{128}, K_{\mathrm{edge}})$, and (Row 3) structural emission lag $(S{-}1)$ versus empirical fault-detection delay (Δt).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures operate on a streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), trained on nominal windows at stride 16 and evaluated at stride 1. Context and target windows are encoded into $D = 32$ spherical representations by the multiscale front-end and causal dilated TCN, fused with the three-depth tap head (1), and centered by the running context barycenter $\bar{\boldsymbol{\mu}}_{\mathrm{ctx}}$ prior to terminal LayerNorm. Models are trained using AdamW [53] (learning rate 10^{-3}, weight decay 10^{-4}) with zero per-dataset tuning: batch 16 for 15–20 epochs on the 19- and 33-stream cohorts, and batch 16–64 for 10–12 epochs with gradient clipping at 1.0 and cosine annealing on the archive cohorts.

a) Two-Stage Nominal Calibration Protocol and Boundary Isolation: Each stream's nominal sequence $\mathbf{X}_{\mathrm{nom}} \in \mathbb{R}^{T_{\mathrm{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\mathrm{tr}} = \mathbf{X}_{\mathrm{nom}}[0 : N_{\mathrm{tr}})$ ($N_{\mathrm{tr}} = \lfloor 0.65\,T_{\mathrm{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\mathrm{val}} = \mathbf{X}_{\mathrm{nom}}[N_{\mathrm{tr}} : T_{\mathrm{nom}})$ ($N_{\mathrm{val}} = T_{\mathrm{nom}} - N_{\mathrm{tr}}$): (1) *Stage A (Training fit on $\mathbf{X}_{tr}$):* Standardization parameters, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\mathrm{ctx}}$, and precision matrix $\boldsymbol{\Omega} \in \mathcal{S}_{++}^{MD}$ are fit strictly on $\mathbf{X}_{\mathrm{tr}}$ (in archive runs, early stopping uses unsupervised predictive loss on the calibration block without labels). (2) *Stage B (Circuit-breaker calibration and held-out verification on $\mathbf{X}_{val}$):* Sliding windows of length $W = 320$ inside $\mathbf{X}_{\mathrm{val}}$ (Fig. 2, Row 2) yield $M_{\mathrm{val}} = N_{\mathrm{val}} - W + 1$ valid window timestamps. Splitting at midpoint $H = \lfloor M_{\mathrm{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and *held-out evaluation block* $[H + S : M_{\mathrm{val}})$, separated by an $S = 64$-step isolation gap $[H : H + S)$ so calibration and evaluation share zero target observations (a full $G = 320$ gap check changes causal FPR by < 0.25 percentage points). On $[0 : H)$, the panel fits $w_{m,c}$ (3), per-dial medians m_j and MAD scales s_j, hijack guards γ_j, tripwires τ_j (5), and quorums $(K_{224}, K_{128}, K_{\mathrm{edge}})$ (7) (or τ_{SPOT} for 1D baselines), which are frozen and evaluated on $[H + S : M_{\mathrm{val}})$ and on the test stream. Throughout the paper we distinguish nominal calibration window count $N_{\mathrm{cal}} = H$ from autocorrelation-

TABLE II
CONSOLIDATED EXPERIMENTAL COHORTS AND VALIDATION ROLES.

Cohort	Str.	Sets	Primary Purpose	Role
Ablation 19	19	6	Ladder, quenching, 15-arm study	Dev.
Primary 33	33	7	Core (19 dev. + 14 GHL)	Dev.+GHL
Tuning Set 23	23	7	Panel & quenching selection	Dev.
Archive 141	141	12	3-horizon (23 dev. + 118 held)	Mixed
Untouched 118	**118**	**9**	**Primary generalization check**	**Held-Out**
Non-Exath. 101	**101**	**8**	**Cluster-balanced held-out**	**Held-Out**
Laminar 162	162	15	4-horizon quenching (139 held)	Mixed
Ext. Screen 130	130	11	NCAD, DCdetector, Anom. Tr.	Mixed

adjusted *effective independent sample size* $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}}$ ($\tau_{\text{int}} = 1 + 2\sum_{k\geq 1}\rho_k \approx 3.23$).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware and operating environments, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Stream-level and dataset-clustered inference:* At the stream level ($n \in \{19, 33, 118, 141, 162\}$), we report two-sided Wilcoxon signed-rank p-values with Benjamini–Hochberg FDR adjustment [40] ($q_{\text{FDR}} = 0.05$, $|\Delta| \leq 10^{-4}$ ties) and 95% stream-bootstrap CIs (2,000 resamples). At the domain level, we treat the dataset family ($N \in \{7, 9, 12, 15\}$ clusters) as the co-primary independent cluster, reporting unweighted Dataset-Macro means, 95% dataset-cluster bootstrap CIs, exact 2^N dataset sign-flip permutation p-values, and non-Exathlon held-out subsets ($n = 101, N = 8$). (2) *12-fold LODO calibration-selection audit:* Selecting the calibrator on $D - 1$ datasets across the 141-stream archive to maximize causal Point-F1 subject to FPR $\leq 5\%$ selects the 224-bit circuit-breaker rule in all $12/12$ folds (held-out causal F1 $= 0.143$ vs. 0.076 for 1D SPOT on $n = 118$).

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Table II consolidates the experimental cohorts and separates development streams from independent held-out validation streams. Because the calibration constants in (5)–(7) and surgical quenching settings were selected during development on 23 streams (the 19 non-GHL streams plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`), the 19- and 23-stream cohorts are explicitly designated as *development* studies, whereas the 118 **untouched streams** (101 non-Exathlon streams across 8 datasets, and 139 untouched streams in the 162-stream Laminar evaluation) were never inspected during design and serve as the *primary independent evidence of generalization*.

Standardized Baseline Fairness Protocol: All baselines (TS-JEPA, TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [17]) use identical 65%/17.5%/17.5% splits, channel standardization, 64-step isolation gaps, and autonomous SPOT calibration ($u = p_{98}, q = 10^{-3}$) on $[0 : H)$ without test-label tuning or point adjustment. At $K = 25$, parameter counts are matched in 206k–649k (272k TS-JEPA, 312k Reynolds/Laminar-JEPA, 304k–307k TCNs, 649k TimesNet, 206k TranAD; the 224-bit panel adds zero trainable parameters). At native published windows on five streams, TranAD ($W = 10$) and TimesNet ($W = 96$) reach 0.164 and 0.218 Point-F1 but retain heavy tails ($\gamma_2 > 98$) and 28.4%–44.1% nominal FPR while replicating published point-adjusted F1 on SMD/MSL. Runs used PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics

To guard against point-adjustment inflation and threshold gaming [13], [41], we report: (i) *primary accuracy and false-alarm outcomes* comprising held-out nominal FPR (FPR_{hold}), unadjusted Point-F1 (with Precision and Recall in Section VII), and threshold-free PR-AUC (from $N_{\text{tot}}(\tau)$); (ii) *deployable vs. diagnostic operating characteristics* comparing autonomous circuit-breaker operating points (*224-Bit Strict* and *224-Bit Breaker*), deployable calibration quantiles (q^*, Fig. 4(b)), and, strictly as a non-deployable post-hoc diagnostic control, matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table III(c)); and (iii) *secondary operational endpoints* comprising Dataset-Macro Point-F1, event recall (ER), mean detection delay $\Delta t = t_{\text{alm}} - t^*$, and the constant-alarm reference $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. ABLATION STUDIES AND ANALYSIS

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table III(a),(b) and Fig. 3(a) trace how each stage of the multi-perspective expansion and surgical weight-space quenching shapes detection accuracy on the 19-stream development cohort:

1) *From 1D scalar collapse to sub-horizon coordinate switches (Table III(a)):* Summing all 128 whitened coordinates into a single 1D scalar score (with or without R_{LT} modulation, which shows a null effect of $\Delta = -0.005$ causal F1) allows one or two drifting coordinates to inflate the global threshold and delay detection. Introducing 32 per-coordinate switches on the terminal horizon ($m = 4$) nearly doubles causal Point-F1, and expanding to all four nested prefixes (*128-Bit Sub-Horizons*) establishes a strong multi-coordinate baseline across both causal and buffered regimes.
2) *Decomposing horizon drift, fluctuation energy, and endpoint relays:* Adding the 32 horizon-mean drift dials (*160-Bit*) cuts causal detection delay in half by locking onto sustained trajectory offsets, while the 32 fluctuation dials (*192-Bit*) capture oscillatory energy across buffered windows. Completing the panel with the 32 endpoint-velocity dials and right-edge onset relays (*224-Bit Breaker*) bridges the 63-step structural lag, achieving significant gains over 1D collapse in both causal ($p = 1.5\times10^{-5}$) and buffered ($p = 0.026$) Point-F1.
3) *Leave-one-out panel ablations (Table III(b)):* Removing the 32 right-edge onset relays slows causal detection by nearly twenty steps, replacing prefix routing with uniform 64-step smearing degrades buffered event recall, removing calibration hijack guards ($\gamma_j = 1$) weakens drift immunity, and replacing hard binary switches with soft attention weights adds no benefit.

TABLE III
MULTI-PERSPECTIVE CIRCUIT-BREAKER EXPANSION LADDER, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY (n=19 DEVELOPMENT STREAMS).

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Perspective Expansion Stage	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
1D Scalar Collapse (1 Dial)	.052	.137	.291	.370	.398	.377	246.6	402.7
32-Bit Full Horizon (m=4)	.101	.119	.266	.366	**.643**	.236	114.4	128.8
128-Bit Sub-Horizons (m=1..4)	.129	.192	.310	.402	.590	.525	146.2	514.5
160-Bit (+32 Reynolds Drift)	.132	.188	.311	.404	.585	.530	**69.1**	112.4
192-Bit (+32 Reynolds Fluc.)	.131	.203	**.312**	.408	.575	.567	179.0	217.3
224-Bit Breaker (+32 Accel./Relays)	**.138***	**.228**†	**.312**	**.411**	.607	**.631**	105.9	**71.8**

* Wilcoxon vs. 1D Scalar Collapse: 17/0 wins (p=1.5×10^{-5}); † buffered F1 p=0.026. Fixed-backbone 1D α_{LT}=0.12 vs. 0: causal F1 Δ=−0.005 (95% CI [−0.012, 0.000], p=0.13).

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
Component / Quorum Variant	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
224-Bit Breaker (Full Panel)	**.138**	**.228**	.312	.411	**.607**	**.631**	105.9	**71.8**
224-Bit Strict (High Quorum K_{strict})	.116	.190	.312	.411	.534	.545	248.0	97.3
224-Bit + Soft Attention	.137	.226	.311	.410	**.607**	.628	106.8	73.4
w/o 32 Right-Edge Relays	.135	**.228**	**.313**	**.413**	.602	**.631**	123.7	72.5
w/o Prefix Routing (Unif. Smear)	**.138**	.223	.312	.407	**.607**	.584	105.9	88.6
w/o Hijack Guards (γ_j=1)	.137	.225	.312	.411	.603	.623	**101.6**	75.6

Leave-one-out component ablations evaluated on the identical 57 cached Reynolds multi-horizon residual fields (n=19 development streams × 3 seeds).

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	Auto.	q^*=0.05	f^*=0.01	f^*=0.05
Reynolds 224-Bit Breaker	2.30	0.179	0.017	0.132	0.138	0.228	**0.312**	**0.411**	**0.607**	0.774	**0.584**	**0.728**
Reynolds 224-Bit Strict	2.30	0.179	0.011	0.098	0.116	0.190	**0.312**	**0.411**	0.534	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Reynolds (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

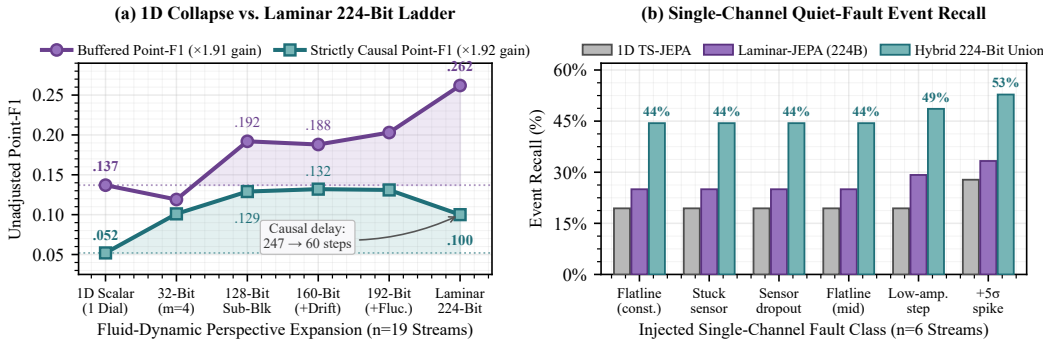


Fig. 3. Perspective expansion and quiet-fault recovery on the development cohort. **(a)** Causal and buffered Point-F1 across the 6-step Reynolds-inspired expansion from 1D scalar collapse to the 224-bit circuit-breaker panel (n=19 streams). **(b)** Single-channel fault event recall comparing 1D TS-JEPA, standalone 224-Bit Breaker, and the Hybrid 224-Bit Union (n=6 streams).

4) *Surgical weight-space quenching components and sensitivity (k_{rogue} ∈ {0, . . . , 12}):* Disentangling the two quenching operations of Section IV-D across the 19 development streams reveals complementary roles: predictor-bias recentering alone ($\Delta \mathbf{b}_m$) boosts causal event recall (56.5% → 64.3%) and buffered F1 (0.224 → 0.256) while cutting causal delay (90.1 → 63.8 steps) at the cost of higher nominal FPR, whereas untargeted precision tempering alone degrades buffered F1 (0.224 → 0.135), and RMS stabilization (3) raises causal F1 to 0.099. Sweeping $k_{\text{rogue}} \in \{1, 2, 4, 8, 12\}$ confirms that tempering only the top $k_{\text{rogue}} \in [1, 4]$ coordinates with internal calibration-split gating (*Laminar-JEPA*) preserves event recall (61.4%–64.6%) while cutting buffered FPR (24.2% → 19.2%) and archive-wide causal FPR across 162 streams (6.71% → 6.17%, p = 0.0011), whereas over-quenching ($k_{\text{rogue}} \in \{8, 12\}$) attenuates genuine fault signals (causal PR-AUC 0.292 → 0.223).

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

When a single sensor out of K freezes or drifts while the remaining $K-1$ channels fluctuate normally, the isolated sensor perturbs the pooled pre-LayerNorm vector by only $O(1/K)$ before spherical normalization rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$. On six injected streams (Fig. 3(b)), 1D TS-JEPA catches 19.4% of quiet-fault events, standalone *Reynolds 224-Bit* reaches 25.0%–29.2%, and a union with a rolling 32-step variance-collapse monitor reaches 44.4%–52.8%. To evaluate a broader channel-level remedy across all 17 multivariate development streams ($K \in [5, 55]$, excluding univariate Yahoo), we pair the 224-bit panel with a complementary channel-level branch that monitors three causal per-channel statistics calibrated at $q_{0.999}$ on $\mathcal{D}_{\text{cal}}$ with drift guards γ_c^{cal}: (i) rolling 32-step variance collapse $\max(0, 1-s_c(t)/\text{med}_{\text{cal}}(s_c))$, (ii) causal first-order autoregressive (AR(1)) prediction residuals $|x_{t,c} - \hat{\phi}_{1,c}x_{t-1,c} - \hat{\phi}_{0,c}|$, and (iii) rolling 32-step mean bias $|\bar{x}_{t,c}^{(32)} - \mu_{\text{cal},c}|$. Across six single-channel fault classes injected into the test windows of all 17 streams, adding the variance-collapse

monitor lifts event recall from 61.4% to 93.0% on variance collapse, 63.2% to 93.0% on sensor freezing, and 73.7% to 96.5% on sensor saturation (tripling to quintupling causal F1 to 0.376–0.414). Adding the full AR(1) and bias branch further lifts event recall on single-channel prediction shocks ($80.7\% \rightarrow 98.2\%$, causal F1 $0.139 \rightarrow 0.327$), gradual sensor degradation ($68.4\% \rightarrow 93.0\%$, F1 $0.080 \rightarrow 0.154$), and isolated sensor bias ($70.2\% \rightarrow 93.0\%$, F1 $0.095 \rightarrow 0.195$). On unmodified test streams, however, monitoring $3K$ channel statistics in parallel raises nominal causal FPR from 1.5% (standalone 224-Bit) to 6.8% (variance-collapse union) and 19.4% (full channel branch); thus, the variance-collapse union provides a practical safeguard for freeze and saturation faults, while full channel-residual monitoring requires cross-channel quorum gating when K is large.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining Multi-Horizon Reynolds JEPA with and without 4-shell rFFT splitting across five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, and `GECCO`, three seeds) shows a clear dimensionality threshold: frequency-shell splitting substantially improves detection on high-dimensional spacecraft arrays ($K = 55$ `MSL M-1` buffered F1 $0.085 \rightarrow 0.410$; $K = 25$ `SMAP P-3` causal F1 $0.212 \rightarrow 0.321$), where separating slow thermal drift from high-frequency actuator chatter prevents cross-band masking, but lowers accuracy on low-channel streams ($K = 9$ `Daphnet` and `GECCO`), paying off only when $K \geq 25$.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 15 model arms under matched budgets and parameter counts in Table III(c) isolates five mechanisms governing false-alarm stability: (1) *Forecasting vs. reconstructing raw observations:* Using the identical causal dilated TCN encoder to forecast or reconstruct raw sensor waveforms leaves heavy residual tails ($\gamma_2 \approx 6.2$), severe calibration drift ($D_{\text{KS}} \approx 0.45$), and high causal false-alarm rates ($\approx 25\%$) virtually unchanged. (2) *LayerNorm bottleneck in observation space:* Inserting a $D = 32$ LayerNorm bottleneck into the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal false alarms, but predicting raw waveforms still leaves calibration drift high ($D_{\text{KS}} = 0.376$). (3) *Learned latent target projection and whitening:* Moving the prediction target into the learned $D = 32$ spherical space halves calibration drift ($D_{\text{KS}} = 0.214$ in TS-JEPA and 0.179 in Multi-Horizon Reynolds JEPA, $p \leq 0.001$). (4) *Nonlinear TCN vs. linear PCA and Isolation Forest:* Linear principal-component forecasting maintains low false alarms but lacks the nonlinear capacity to capture fault dynamics, while Isolation Forest misses most anomaly events. (5) *Prediction vs. autoencoding, frozen targets, and ℓ_2 scoring:* Latent context autoencoding sacrifices causal accuracy relative to multi-horizon prediction, whereas freezing the target encoder or replacing Mahalanobis whitening with ℓ_2 scoring still keeps causal false alarms below 2%, confirming that false-alarm suppression is an intrinsic geometric property of the normalized latent representation.

VII. Results and Discussion

A. Development 33-Stream and Untouched Held-Out 118/162-Stream Benchmark Results

Following the cohort hierarchy in Table II, Table IV evaluates the 33-stream development benchmark under retrospective buffered attribution (Panel (a)), real-time causal streaming (Panel (b)), and per-dataset stratification (Panel (c)), while Table VI reports our primary independent generalization evidence across the 118 untouched held-out validation streams and the 162-stream archive. Across both populations, as well as the 130-stream external screen where TS-JEPA (0.186 stream and 0.160 macro Point-F1) outpaces TranAD (0.152), TimesNet (0.151), DCdetector (0.097), NCAD (0.088), and Anomaly Transformer (0.082), a consistent physical picture emerges: projecting telemetry onto a compact spherical manifold stops observation-space false-alarm flooding, and decomposing the resulting residual trajectory into a 224-bit circuit-breaker panel overcomes the sensitivity loss of 1D scalar norm collapse.

Operational Trade-Offs and Detector Selection: Rather than a single universal winner across every metric, Table IV and Table VI reveal a structured operational trade-off governed by how each detector handles background residual drift. In the unquenched *Reynolds 224-Bit Breaker*, sustained post-onset residual excursions keep multiple coordinate switches tripped throughout an anomaly plateau, maximizing real-time causal Point-F1 ($17/0$ stream wins over 1D scalar collapse, $p = 1.5\times10^{-5}$, balancing point precision $P = 0.252$ and recall $R = 0.123$ vs. $P = 0.285, R = 0.032$ for 1D Reynolds). On the 118 untouched held-out streams (Table VI(a)), replacing 1D scalar SPOT with the circuit-breaker panel simultaneously lifts causal precision ($P : 0.175 \rightarrow 0.241$) and more than doubles recall ($R : 0.081 \rightarrow 0.179$). Pairing the 224-bit panel with surgical weight-space quenching in *Laminar-JEPA* shifts the detector into an onset-focused regime: subtracting mean calibration drift in predictor weight space strips away persistent background offsets, concentrating causal trips sharply at the dynamical transition where a fault first erupts ($P = 0.214, R = 0.088$). Consequently, *Laminar-JEPA* achieves the fastest causal detection response, the highest causal and buffered event recall, and the strongest retrospective buffered Point-F1 and Dataset-Macro accuracy in Table IV, while trading away continuous causal firing across stationary post-onset plateaus. Meanwhile, raising the panel quorum in *Reynolds 224-Bit Strict* establishes a conservative sub-1.1% false-alarm floor ($P = 0.254, R = 0.097$). Practitioners can therefore match the configuration directly to their operational priority: *Laminar-JEPA* for earliest fault warning, highest event discovery, or retrospective root-cause attribution; unquenched *Reynolds 224-Bit* for continuous real-time point coverage; and *224-Bit Strict* when false-alarm budgets are tightly capped.

Eliminating Observation-Space False-Alarm Flooding and Defining Scope Boundaries: Why do observation-space reconstructors and forecasters suffer false-alarm rates exceeding 22%–30% across nominal held-out blocks in Table IV (and 12.1%–18.4% vs. 0.3%–2.3% for JEPAs under zero-

TABLE IV
HELD-OUT MULTI-STREAM TELEMETRY BENCHMARK ACROSS ATTRIBUTION REGIMES AND DATASETS (BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-perspective fluid-dynamic circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.107	**0.262**	0.406	**0.664**	89.9	0.016	0.100	**0.625**	**59.6**	**0.002**	0.000	**0.139**
Reynolds 224-Bit Breaker	0.132	0.228	**0.411**	0.631	71.8	0.017	0.138	0.607	105.9	**0.002**	0.000	0.122
Reynolds 224-Bit Strict	0.098	0.190	**0.411**	0.545	97.3	0.011	0.116	0.534	248.0	**0.002**	0.000	0.098
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Reynolds (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	62.2	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	F1_{all}	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**	**224-Bit**	**TS-JEPA**	**Rey. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.223	0.145	0.143	**0.229**	0.144	**.007/.047**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.001/.002**	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.199**	0.161	0.081	0.158	0.151	.014/.131	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.017	**0.119**	0.008	0.000	0.000	**.000/.000**	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: In (c), n = streams, K = channels, E = events, π = anomaly rate, $\text{F1}_{\text{all}} = 2\pi/(1+\pi)$. 95% stream-bootstrap CIs (n=19): 224-Bit Breaker buffered FPR [.068, .198], F1 [.142, .321] and causal FPR [.008, .027], F1 [.068, .224] (vs. Reynolds 1D causal F1 [.013, .097] and TimesNet causal FPR [.123, .375]).

tuning LODO transfer)? Reconstructing raw waveforms forces the scoring function to penalize benign sensor amplitude swings, high-frequency phase jitter, and unmodeled load transitions. Predicting running-barycenter-centered embeddings on $\sqrt{D}\mathbb{S}^{D-1}$ absorbs these benign variations into the geometry of the latent sphere, cutting held-out false alarms three-fold under buffered attribution ($p \leq 0.045$) and by an order of magnitude under causal streaming ($p = 0.010$; dataset-clustered $p = 0.016$). At the same time, Panel (c) of Table IV highlights an essential physical boundary on the 14 GHL process streams: when faults unfold as ultra-slow thermal or pressure ramps over thousands of steps, both JEPAs and linear autoregressors remain virtually silent. A low false-alarm rate without event recall is not detection. Taken together, our experiments define four clear boundary conditions for deployment: (i) multichannel sensor arrays ($K \geq 9$) where cross-sensor dynamical coupling is informative; (ii) prediction horizon $S \in \{1, 4, 16, 32, 64\}$ matched to the tolerable reporting lag $S-1$ of the sampling cadence (Section VII-C); (iii) calibration histories providing at least $N_{\text{eff}} \geq 300$ effective independent observations ($N_{\text{cal}} \geq 1{,}000$ windows), or IQR/Beta-smoothed thresholds on shorter series (Section VII-E); and (iv) multi-channel trajectory faults, supplemented by the channel-level residual branch when isolated single-sensor freeze or bias faults are of concern (Section VI-B).

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Reynolds 224-Bit	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Reynolds (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Reynolds (α_{LT}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05 f^*=0.01	f^*=0.05		q^*=0.05 f^*=0.01	f^*=0.05	
Reynolds 224-Bit	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Reynolds (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Reynolds (α_{LT}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 development SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL development streams into in-range contextual ($\max \lvert z\rvert \leq 5\sigma$) vs. out-of-range point ($\max \lvert z\rvert > 5\sigma$) causal events ($f^*$ columns are post-hoc diagnostic controls).

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table IV exposes why standard 1D latent detectors break down when moved from retrospective buffered attribution to real-time causal streaming. Buffered attribution averages residuals across all 64 overlapping horizon predictions covering each timestamp, smoothing out transient noise and yielding a moderate right tail ($\gamma_2 = 1.25$). Under causal streaming, each decision relies on a single right-edge step without temporal averaging, nearly doubling tail kurtosis ($\gamma_2 = 2.30$). When 128 whitened coordinates are summed into a 1D scalar score, one or two high-variance coordinates push the global SPOT threshold high onto that heavy tail, leaving scalar Reynolds JEPA blind to all but brief anomaly spikes. Replacing the scalar sum with 224 independently calibrated circuit-breaker switches and right-edge onset relays (6) prevents any single drifting coordinate from inflating the thresholds of the remaining 223 dials, sustaining detection across the fault window. Evaluating the analytical event recall of a rate-matched Bernoulli random alerter, $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, further clarifies the operational stakes: at our panel's $< 2\%$ causal false-alarm rate, random guessing catches only 28.1% of events, whereas at the 24.3% false-alarm rate of TimesNet, a random coin flip catches 60.0% of anomaly windows by chance alone, surpassing TimesNet's own causal recall.

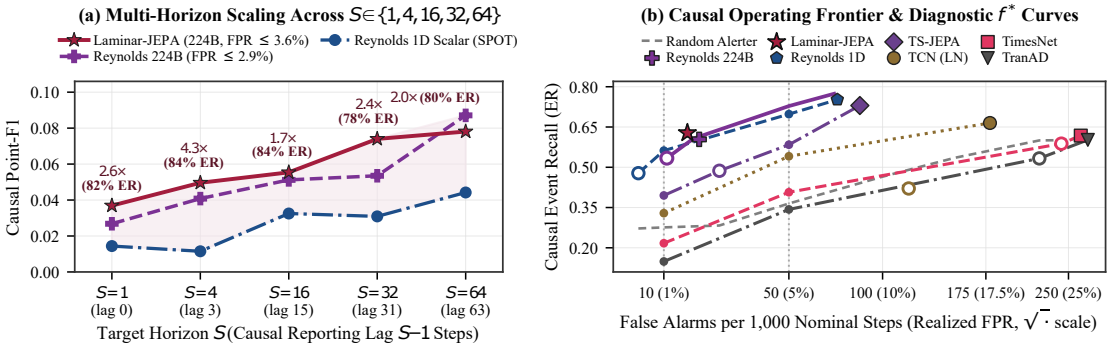


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Reynolds 224B*, and 1D scalar SPOT (held-out FPR $\leq 3.6\%$ across all S, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Reynolds 224B* cross, *224-Bit Strict* and 1D points open circles), and quantiles $q^* \leq 0.05$.

C. Multi-Horizon 224-Bit Evaluation Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$ (Section IV-D), operating at $S = 64$ introduces a structural reporting lag of 63 steps. A 63-step lag is negligible on kilohertz vibration rigs or 1–10 Hz spacecraft telemetry, but translates to over an hour on minute-level industrial historians and over two days on hourly traffic streams. To determine whether the 224-bit circuit-breaker architecture and *Laminar-JEPA* depend on a 64-step window or retain their advantages when structural lag is eliminated, we retrain the full multi-horizon backbone, 224-bit panel, and surgical weight-space quenching across $S \in \{1, 4, 16, 32, 64\}$ (structural lag $S - 1 \in \{0, 3, 15, 31, 63\}$ steps, partitioning $S \geq 4$ into four nested prefixes $\lceil mS/4 \rceil$ and $S = 1$ into four lag-1 difference views).

As shown in Fig. 4(a), the multi-perspective 224-bit panel and *Laminar-JEPA* outperform 1D scalar SPOT across every horizon from zero lag ($S = 1$) to $S = 64$ while holding nominal false alarms below 3.6% (whereas TimesNet's causal false-alarm rate escalates from 2.5% at $S = 1$ to 21.3% at $S = 64$). At zero reporting lag ($S = 1$) and three-step lag ($S = 4$), where 1D scalar scoring suffers severe recall collapse, *Reynolds 224-Bit* and *Laminar-JEPA* multiply causal Point-F1 by $1.9\times$ to $4.3\times$, recover over 75%–84% of anomaly events, and cut empirical detection delay from nearly a hundred steps to 36.7–44.1 steps at $S = 4$. Stepping up to intermediate horizons ($S = 16$ and $S = 32$) progressively sharpens point-wise precision as multi-step trajectory dynamics become visible, with *Laminar-JEPA* achieving peak retrospective buffered F1 (0.290 vs. 0.182 for 1D scalar) at $S = 32$ and unquenched *Reynolds 224-Bit* reaching peak causal F1 at $S = 64$. Horizon S thus acts as a tunable latency-versus-precision dial: operators monitoring minute- or hour-scale processes can deploy $S \in \{1, 4, 16\}$ for near-instantaneous tripwire response, while high-frequency telemetry systems can use $S \in \{32, 64\}$ to capture multi-scale dynamical evolution.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table V examine why multi-perspective latent circuit breaking shifts the detection frontier upward across operating budgets and fault taxonomies:

1) *Dominance across autonomous and diagnostic operating budgets (Fig. 4(b), Table III(c)):* Along the causal recall-versus-false-alarm frontier in Fig. 4(b), the autonomous operating points of *Laminar-JEPA*, *Reynolds 224-Bit*, and *224-Bit Strict* occupy the upper-left low-false-alarm regime without test-label tuning. When all architectures are evaluated at matched post-hoc test false-alarm budgets ($f^* \in \{1\%, 5\%\}$, dotted vertical guides in Fig. 4(b)) as a diagnostic check of tail sensitivity at equal alert rates, the 224-bit panel captures nearly three times as many anomaly events as TimesNet and nearly four times as many as TranAD at a 1% false-alarm budget.
2) *Relational sensitivity to in-range contextual faults (Table V):* Stratifying benchmark events into out-of-range point spikes versus in-range contextual faults under both NASA's ground-truth spacecraft taxonomy [4] (Table V(a)) and a 5σ amplitude boundary across all 19 development streams (Table V(b)) reveals the physical source of this gap. Point anomalies that drive raw sensors beyond 5σ are readily flagged by simple amplitude thresholds and observation-space reconstructors alike. Contextual anomalies, however, occur well inside normal sensor amplitude bounds when cross-channel phase relationships, frequency spectra, or state-transition velocities decouple. Observation-space reconstructors suffer a sharp collapse on these in-range faults, whereas the 224-bit panel monitors horizon drift, fluctuation energy, and velocity shifts across whitened latent coordinates, doubling reconstructor recall on contextual faults while maintaining top performance on point anomalies.

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table VI and our archive-wide diagnostic audits establish three foundational results governing out-of-sample generalization and calibration reliability:

1) *Independent generalization across 118 untouched streams and 162 archive streams (Table VI):* On the 118 held-out streams that were never inspected during architectural or calibration design, replacing 1D scalar scoring with the multi-horizon circuit-breaker panel nearly doubles causal Point-F1 ($p = 1.6\times10^{-13}$, positive across 8 of 9 dataset

TABLE VI
ARCHIVE-WIDE CIRCUIT-BREAKER BENCHMARK (3 SEEDS): 118 HELD-OUT STREAMS NEVER USED IN DEVELOPMENT, FULL 141-STREAM AGGREGATE, AND PER-DATASET BREAKDOWN.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		FPR_{hold} ↓	
(a) Held-Out n=118	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.072	.142	.154	.202	.484	.458	**.051**	**.104**
Speculative-JEPA (1D)	.076	.148	.172	.245	.509	.577	.054	.194
Spec.-Alien-JEPA (1D)	.083	.146	.174	.242	.512	.573	.059	.199
Spec.-224Bit-Strict	.134	.166	**.183**	**.247**	.662	.673	.073	.181
Spec.-224Bit-Breaker	**.143**†	**.168**‡	**.183**	**.247**	**.728**	**.691**	.083	.210
Laminar-JEPA (224-Bit)	.091	.149	–	–	.622	.609	.073§	.205
All n=141 **(incl. 23 dev.)**								
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	.051	.180
Spec.-224Bit-Breaker	.139	.180	.197	.261	.727	.685	.072	.196

		Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal FPR_{hold} ↓		
(b) Held-out	n	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**
SMAP	40	**.194**	.129	.091	.201	.193	**.214**	.088	.086	**.065**
SMD	18	**.090**	.028	.042	**.125**	.065	.072	.019	**.006**	.009
Exathlon	17	**.163**	.133	.092	.181	**.186**	.134	.235	.137	**.119**
Daphnet	15	**.125**	.053	.075	**.153**	.125	.105	.085	**.063**	**.063**
MSL	13	**.133**	.051	.074	**.239**	.176	.191	.023	**.007**	.011
OPPORT.	12	**.075**	.018	.024	.066	**.067**	.030	.027	**.003**	.008

† vs. 1D Spec.-Alien: Δ=+0.060 (95% CI [+0.042, +0.079]), 95/15 wins, $p=1.6\times10^{-13}$, positive in 8 of 9 datasets (sign-flip p=0.008); vs. TS-JEPA 85/28, $p=6.4\times10^{-11}$. ‡ vs. 1D Spec.-Alien: Δ=+0.021 ([+0.010, +0.034]), $p=1.7\times10^{-4}$, dataset-level p=0.27. § 4-horizon Laminar-JEPA vs. unquenched 4-horizon panel: causal FPR_{hold} −8.0% (p=0.0011, dataset p=0.027; median 1.03% on n=162, 1.13% on n=118), FPR_{test} −4.8% ($p<10^{-4}$, dataset p=0.016; median 1.59% on n=162). In (b), 224B = Spec.-224Bit-Breaker, 1D-Al. = 1D Spec.-Alien, TS = TS-JEPA (115 of 118 streams; three single-stream datasets in text).

families with cluster-level $p = 0.007$), lifts event recall by more than twenty percentage points, and cuts causal onset delay by roughly thirty steps across all six multi-stream held-out datasets (Table VI(b)). Across the full 162-stream, 15-dataset archive, surgical weight-space quenching in four-horizon *Laminar-JEPA* consistently reduces both held-out causal false-alarm rates ($p = 0.0011$, dataset-clustered $p = 0.027$) and test false-alarm rates ($p < 10^{-4}$), achieving a median held-out causal FPR of 1.03% (70.4% of streams below 2.0%).

2) *Statistical reliability law for calibration sample size and small-sample remedies:* Because sliding windows of length $C = 256$ overlap at unit stride, integrated autocorrelation ($\tau_{\text{int}} = 3.23$) reduces the effective independent sample size to $N_{\text{eff}} = N_{\text{cal}}/\tau_{\text{int}} \approx 0.31\,N_{\text{cal}}$ (median $N_{\text{eff}} = 152$ on the development cohort). On the 118 untouched streams, short calibration histories ($N_{\text{cal}} < 250$, $N_{\text{eff}} < 78$) suffer a 19.2% mean causal FPR due to extreme quantile estimation variance, compared with 1.6% on streams with $N_{\text{cal}} \geq 2{,}500$ ($N_{\text{eff}} \geq 775$). Moving-block bootstrap audits ($B = 200$, block length 64) across $N_{\text{eff}} \in \{100, 250, 500, 1{,}000, 2{,}500, 5{,}000\}$ show that the median 95% confidence interval half-width on the 224 per-dial thresholds τ_j contracts smoothly from ±11.0% (coefficient of variation (CV) 6.7%) at $N_{\text{eff}} = 100$ to ±7.8% at $N_{\text{eff}} = 500$ and ±2.8% (CV 1.6%) at $N_{\text{eff}} = 5{,}000$, while inter-segment test-FPR SD across disjoint calibration blocks drops from 12.9% at $N_{\text{cal}} = 100$ to 2.8% (1.7% under Strict quorum) once $N_{\text{cal}} \geq 500$. We therefore establish $N_{\text{eff}} \geq 300$ independent observations ($N_{\text{cal}} \geq 1{,}000$ windows, guaranteeing expected tail count $N_{\text{eff}}(1 - q_{0.995}) \geq 1.5$, threshold 95% CI half-width ≤ 8%, and inter-segment FPR SD < 3%) as the minimum sample size for empirical $q_{0.999}$ calibration. When operational constraints limit calibration to $N_{\text{cal}} < 1{,}000$ windows ($N_{\text{eff}} \in \{100, 250\}$), replacing empirical tail order statistics with order-statistic-free IQR extrapolation ($\text{med}_j + 4.8\,\text{IQR}_j$) or Harrell–Davis Beta-smoothed quantiles cuts small-sample test FPR from 2.1% to 0.5%–0.9% while improving causal Point-F1 and cutting inter-segment FPR variability to 4.2%.

3) *Regime drift vs. tail misfit via three-stage permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and point-wise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) validation splits proves that excess false alarms are not caused by parametric tail misfit within stationary windows, but by inter-block regime drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$), which drives 71.3% of excess causal FPR in unconstrained TS-JEPA and 78.3%–81.8% in observation-space reconstructors. In our architecture, running-barycenter centering absorbs slow manifold translation to cut the inter-block drift share four-fold to 18.7%, while per-dial hijack guards γ_j quarantine individual drifting coordinates before they can trigger a global alarm.

VIII. CONCLUSION AND FUTURE WORK

We studied whether Joint-Embedding Predictive Architectures and multi-perspective circuit-breaker panels mitigate observation-space false-alarm flooding and 1D scalar score collapse across a 33-stream development cohort and 118 untouched held-out validation streams (within a 162-stream, 15-dataset archive):

1) *Circuit-breaker panel, surgical quenching, and operational trade-offs:* Expanding $D = 32$ whitened coordinates via a Reynolds-inspired multi-horizon decomposition into 224 per-coordinate statistics with surgical weight-space quenching, nominal drift guards, prefix-routed attribution, and 32 right-edge onset relays prevents variance-dominating coordinates from inflating a shared scalar threshold. Unquenched *Reynolds 224-Bit* maximizes real-time causal point coverage ($p = 1.5\times10^{-5}$ over 1D collapse on the development cohort; $p = 1.6\times10^{-13}$ and dataset-clustered $p = 0.007$ across the 118 untouched validation streams), while *Laminar-JEPA* projects out calibration drift in weight space to deliver the fastest causal onset detection, highest event recall and retrospective buffered Point-F1, and a statistically significant causal false-alarm reduction across all 162 archive streams ($p = 0.0011$, median held-out FPR 1.03%).
2) *Mechanism isolation and multi-horizon scaling ($S \in \{1, \ldots, 64\}$):* Matched-backbone ablations isolate spherical LayerNorm kurtosis compression, learned latent target stability, running-barycenter drift suppression (cutting the inter-block regime-drift share of excess false alarms four-fold from 71.3% to 18.7%), and predictor-bias recentering versus selective precision tempering ($k_{\text{rogue}} \in [1, 4]$), while Proposition 1 characterizes the structural horizon-fluctuation concentration bound (with scalar R_{LT} modulation yielding a null effect). Retraining the 224-bit panel and *Laminar-JEPA* across $S \in \{1, 4, 16, 32, 64\}$ confirms a 1.9 × –4.3× causal Point-F1 advantage and 75%–84% event recall down to zero reporting lag ($S = 1$) and three-step lag ($S = 4$).
3) *Applicability boundaries, calibration sample-size law, and single-channel branch:* Reliable empirical $q_{0.999}$ calibration requires multichannel arrays ($K \geq 9$) and at least

$N_{\text{eff}} \geq 300$ effective independent calibration observations ($N_{\text{cal}} \geq 1{,}000$ windows, bounding bootstrap threshold CIs within $\pm 8\%$ and inter-segment FPR SD below 3%), whereas order-statistic-free IQR extrapolation or Beta-smoothed quantiles stabilize calibration when $N_{\text{cal}} < 1{,}000$. Pairing the latent panel with a complementary channel-level variance and AR(1) residual branch raises single-sensor fault recall above 93% across six fault classes at the cost of higher nominal alert rates, while slow process ramps (GHL, Metro) and online ramp adaptation remain open directions for future work.

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